

Exact entropy-production fibers and local upper bounds in partially observed Markov networks

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Complete waiting-time statistics between resolved transitions constrain a hidden Markov network, but need not determine its steady-state entropy production. We characterize an exact entropy-production identified set and its sensitivity to observational error for finite, irreducible, bidirected continuous-time Markov chains with one observed reverse pair. Under a five-state cap, an explicit one-parameter family has unbounded compatible entropy below an algebraic threshold, exactly two distinct finite entropy values at the threshold, and a unique compatible generator above it. A global invariant-triangle argument exhausts the physical realizations, including sparse supports; an independent algebraic construction certifies the two endpoint values. We then prove a sharp local criterion at fixed state count and unknown topology: a finite entropy upper bound holds in some positive neighborhood of the joint waiting laws precisely for two states or a complete three-state source. The criterion survives fixing the observed rates and generator trace. Three positive Laplace arguments recover every three-state generator locally, but entropy can still diverge as a missing reverse pair opens asymmetrically. With two fully hidden states, a complete perturbation followed by an exact observation-preserving similarity produces unbounded entropy in every neighborhood. Thus exact thermodynamic identification and finite local upper bounds impose distinct requirements, even when all microscopic rates are uniformly bounded above.

I. INTRODUCTION

Partial observations of a stochastic network can reveal broken time-reversal symmetry without resolving every dissipative transition. Waiting times between visible jumps retain information about excursions through the hidden network. Recent methods use this information to bound entropy production, infer affinities, or reconstruct aspects of network topology [1–3]. These successes motivate a complementary question: what is the *entire set* of microscopic entropy-production rates compatible with the observation, and does a finite upper bound persist when exact equality is relaxed?

The distinction matters because the relative entropy between an observed stationary path law and the observation of its physical time reverse cannot exceed the microscopic path relative entropy, by data processing [1, 2]. Hidden nonequilibrium can remain compatible with much less asymmetric

observed data [4, 5]. Even complete, infinite-statistics waiting-time laws define an inverse problem over physical generators, with positivity and support constraints. A single fitted generator, a lower bound, or a family of observationally equivalent models does not by itself determine the full entropy identified set. Compatible-model minimization already provides a systematic route to lower bounds [5, 6]; here we ask about the full image and its upper extent.

The mathematical machinery for this inverse problem has substantial precedents. Positive hidden Markov models admit similarity-based equivalence descriptions [7]; related representations occur for Markov arrival processes [8]. Phase-type representations can be studied through invariant polytopes [9], and stochastic matrices can have finite sets of nonnegative factorizations [10]. We use these ideas with fixed, individually resolved microscopic reset transitions and reciprocal physical support. We do not claim a new general realization theory or that finite positive-realization ambiguity is itself new.

Our first result is an explicit, globally exhausted physical entropy fiber. Along a balanced five-state family, its entropy image changes from unbounded to two-point to singleton at a specified quartic root. The two-point assertion counts all admissible generators under the state cap, rather than representatives in a chosen canonical form. Our second result classifies local upper boundedness for every fixed state count with one observed reverse pair. Only two-state chains and complete three-state sources have a finite upper ceiling in a neighborhood of their joint waiting laws. Both positive and negative statements retain their force when the observed rates and trace are fixed.

The mechanism has two parts. At a sparse generator, a vanishing current can multiply a still faster diverging affinity. At larger state count, hidden similarities can reach a one-way limiting boundary while leaving the observed law exactly unchanged. The singularity of relative entropy and the effect of vanishing reverse rates are established phenomena [11, 12]. The contribution here is the explicit physical-fiber classification and the necessary-and-sufficient local criterion under the stated observation model. We first formulate that model and state the exact-fiber result, then derive the local criterion and its finite-record consequence. The appendices contain the full realization, support-exhaustion, construction, and arithmetic arguments.

II. MODEL, OBSERVATIONS, AND IDENTIFIED SETS

Fix a physical time unit and two named states x, y . For $N \geq 2$, let $\mathcal{C}_N$ consist of row generators Q on $\{x, y, h_1, \dots, h_{N-2}\}$ with $Q\mathbf{1} = 0$, nonnegative off-diagonal entries, connected reciprocal

support

$$q_{ij} > 0 \iff q_{ji} > 0, \quad i \neq j, \quad (1)$$

and $r = q_{xy} > 0$, $s = q_{yx} > 0$. Every rate is finite. There is no common rate floor or upper bound unless stated, and competing support graphs are unknown. Set $\mathcal{C}_{\leq m} = \bigcup_{N=2}^m \mathcal{C}_N$. A fixed count and a cap are different assumptions.

There is one kinetic channel per ordered state pair, the generator is time independent, and all states are even under time reversal. The generator entropy-production functional is

$$\sigma(Q) = \sum_{i < j: q_{ij} > 0} (F_{ij} - F_{ji}) \log \frac{F_{ij}}{F_{ji}}, \quad F_{ij} = \pi_i q_{ij}, \quad (2)$$

$$= \sum_{i \neq j: q_{ij} > 0} \pi_i q_{ij} \log \frac{q_{ij}}{q_{ji}}, \quad (3)$$

where $\pi Q = 0$, $\pi \mathbf{1} = 1$ is the unique positive stationary distribution. Equation (3) follows by stationary cancellation of the $\log \pi_i$ terms. We use natural logarithms and $k_B = 1$, so σ has inverse-time units. A paired absent edge contributes zero. Every member of $\mathcal{C}_N$ has finite nonnegative entropy, zero exactly at detailed balance. This is the usual finite-network convention [13, 14]; interpreting it as all physical dissipation additionally requires resolved thermodynamic channels and local detailed balance. Parallel unresolved channels or odd variables require a different model.

Observe only marks $+= (x, y)$ and $- = (y, x)$, with known endpoints. Define

$$E_+ = r e_x e_y^T, \quad E_- = s e_y e_x^T, \quad T = Q - E_+ - E_-, \quad (4)$$

$$R = \begin{pmatrix} e_y^T \\ e_x^T \end{pmatrix}, \quad B = (r e_x, s e_y). \quad (5)$$

Crucially, T retains the diagonals of Q . Killing at a marked jump makes T transient. The observation is the matrix of joint next-mark/time densities

$$\mathcal{O}(Q) = \Psi_Q(t) = R e^{Tt} B, \quad \widehat{\Psi}_Q(\lambda) = R(\lambda I - T)^{-1} B, \quad \lambda > 0. \quad (6)$$

Each row is a probability measure $\mu_I(dt, J) = \psi_{IJ}(t) dt$ on $[0, \infty) \times \{+, -\}$. Individual entries need not integrate to one. Absolute time and joint amplitudes are part of the data. The reset property determines all successive marked-event laws from these rows [1, 2].

Two models are observationally equivalent when their two row laws agree, equivalently when $\Psi_Q(t) = \Psi_{Q'}(t)$ for every $t \geq 0$. Physical models are identified only under permutations of hidden states fixing x, y . For a declared class $\mathcal{C}$, define

$$\mathcal{F}_{\mathcal{C}}(D) = \{[Q]_{\text{lab}} : Q \in \mathcal{C}, \mathcal{O}(Q) = D\}, \quad \mathcal{S}_{\mathcal{C}}(D) = \{\sigma(Q) : [Q]_{\text{lab}} \in \mathcal{F}_{\mathcal{C}}(D)\}. \quad (7)$$

Generator identification means $|\mathcal{F}_\mathcal{C}(D)| = 1$; thermodynamic identification means $|\mathcal{S}_\mathcal{C}(D)| = 1$. A bounded nonunique entropy fiber has more than one value and finite supremum. An unbounded fiber contains arbitrarily large *finite* values; no model with a one-way edge is admitted.

For approximate data we use maximum row total variation,

$$d(D, D') = \max_I \frac{1}{2} \sum_J \int_0^\infty |\psi_{IJ}(t) - \psi'_{IJ}(t)| dt. \quad (8)$$

We also consider the product weak topology of the two row probability measures. A *finite local upper ceiling* at D in $\mathcal{C}$ means that some $\eta > 0$ satisfies

$$\sup\{\sigma(P) : P \in \mathcal{C}, d(\mathcal{O}(P), D) < \eta\} < \infty. \quad (9)$$

Its weak version replaces the ball by an open neighborhood. This property is defined even when exact entropy is multivalued. At an identifiable source, convergence of entropy along all observationally convergent models is stronger than a finite ceiling. Stationary records on a fixed time window form a separate experiment, treated in Sec. VI.

III. AN EXACTLY TWO-POINT ENTROPY FIBER

The following family displays all three exact entropy regimes within a single fixed-dimensional inverse problem. Rates are expressed in the time unit of Sec. II. Linear minimality means that the columns of $T^j B$ span the state space and that $RT^j v = 0$ for every $j \geq 0$ forces $v = 0$. The minimal linear order is the least dimension of any matrix realization of the kernel; physical positivity is an additional constraint.

Theorem 1 (Balanced five-state fiber). *In state order (x, y, h, k, l) , let*

$$Q(z) = \begin{pmatrix} -11 & 1 & 2 & 0 & 8 \\ 2 & -20 & 0 & 7 & 11 \\ 3 & 0 & -7 & 4 & 0 \\ 0 & 6 & 5 & -11 & 0 \\ z & z & 0 & 0 & -2z \end{pmatrix}, \quad z > 0. \quad (10)$$

Let z_ be the unique positive root of*

$$\mathcal{P}(z) = 352z^4 - 3168z^3 - 4644z^2 - 11340z - 7623. \quad (11)$$

Every $\Psi_{Q(z)}$ has minimal linear realization dimension five. In $\mathcal{C}_{\leq 5}$ its entropy fiber is unbounded for $0 < z < z_$, contains exactly two distinct finite values at $z = z_*$, and is a singleton with a unique*

physical generator for $z > z_*$. At z_* , the two generators are $Q(z_*)$ and the algebraic generator Q_* defined in Appendix C. Their entropy values satisfy the rational enclosures

$$0.062946400236 \leq \sigma(Q(z_*)) \leq 0.062946400237, \quad (12)$$

$$0.065579685966 \leq \sigma(Q_*) \leq 0.065579685967. \quad (13)$$

Here $z_* = 10.5571496686650 \dots$ is an illustrative decimal approximation.

The assertion at z_* is stronger than exhibiting two observationally equivalent models. It excludes every third physical generator under the cap, including any on a different bidirected support. The source has six reciprocal edges; the alternative has seven and a connected hidden path (Fig. 1). Thus hidden relabeling cannot explain their difference. Above the threshold even the physical generator is unique, although its observation will have no finite local entropy upper ceiling by Theorem 4 below.

Proof structure. The marked kernel fixes the visible block and the hidden transfer $X(\lambda I - H)^{-1}Y$ for the partition $Q = \begin{pmatrix} A & X \\ Y & H \end{pmatrix}$. Minimality and the state cap force every competitor into a normalized hidden-similarity orbit; Appendix A proves this reduction. A disconnected hidden support

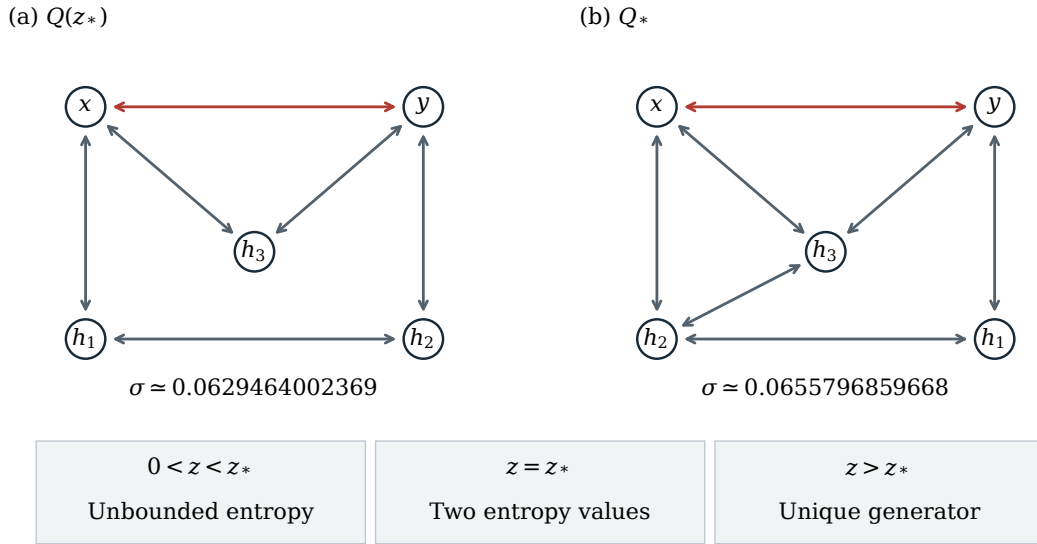


FIG. 1. Exact classification for the family (10) under a five-state cap. The strip denotes properties of the entire entropy fiber, not a numerically optimized maximum. At the algebraic threshold the source and the connected-hidden alternative are the only physical realizations up to hidden labels. Paired arrows denote reciprocal support, not equal rates; only $x \leftrightarrow y$ is observed. Entropy values in the figure are rounded; Eqs. (12)–(13) give certified intervals.

can realize the fast-parameter data only as the source, by a residue-sign and reciprocity argument (Appendix B). Every connected hidden support gives a finite invariant triangle after Perron normalization. Global containment and inward-flow inequalities reduce feasibility to a quadratic whose discriminant is a negative positive-factor multiple of $\mathcal{P}(z)$. Above z_* this excludes every connected realization. At z_* , equality forces a single triangle and hence at most one normalized generator. Appendix C independently constructs that generator, verifies physical feasibility, and separates its entropy from the source using rigorous logarithm bounds. Finally, Appendix E covers all $0 < z < z_*$ by exact positive realizations with a hidden pair whose entropy-divergent similarity orbit is proved in Appendix D. These arguments establish the whole fiber, not a count from a finite search. $\square$

The cap has physical content. It excludes additional nonminimal states invisible to the linear realization. Splitting one fully hidden state of (10) into two strongly lumpable copies preserves the complete marked kernel and permits entropy divergence on their internal edge (Appendix F). Thus one additional state makes the endpoint entropy fiber unbounded. Conversely, the cap can be replaced by requiring all competing realizations to be linearly minimal. No separate upper rate cap is needed in the exact fiber: similarity fixes $-\text{tr } Q = 49 + 2z$, which bounds each nonnegative off-diagonal rate. We make no general classification claim for arbitrary five- or six-state data.

IV. STABLE GENERATOR RECOVERY AND SINGULAR ENTROPY

Three states isolate the difference between recovering rates and recovering entropy. There is only one hidden state, so no nontrivial hidden similarity remains, but this algebraic observation alone would not establish stability.

Theorem 2 (Three positive Laplace arguments). *For every $Q_0 \in \mathcal{C}_3$ and any $0 < \lambda_1 < \lambda_2 < \lambda_3$, the three matrices $\widehat{\Psi}_{Q_0}(\lambda_j)$ determine Q_0 uniquely within $\mathcal{C}_{\leq 3}$. On admissible data near these matrices, the inverse is rational with nonzero denominators and locally Lipschitz. Consequently weak convergence of joint row laws to $\mathcal{O}(Q_0)$ among models in $\mathcal{C}_{\leq 3}$ implies eventual dimension three and entrywise convergence to Q_0 . In row TV the inverse is locally Lipschitz. At a complete three-state source, entropy is also locally Lipschitz in row TV and continuous under weak row convergence.*

Proof. Write

$$Q = \begin{pmatrix} -r-a & r & a \\ s & -s-c & c \\ b & d & -b-d \end{pmatrix}, \quad h = b + d > 0. \quad (14)$$

Reciprocity and irreducibility imply $a + c > 0$. Let P interchange the two mark rows, $F(\lambda) = P\widehat{\Psi}(\lambda)$, and $M(\lambda) = F(\lambda)^{-1}$. The visible resolvent K satisfies $F = K \operatorname{diag}(r, s)$ and

$$K^{-1} = \operatorname{diag}(\lambda + r + a, \lambda + s + c) - \frac{1}{\lambda + h} \begin{pmatrix} a \\ c \end{pmatrix} (b, d). \quad (15)$$

This is a nonsingular strictly row diagonally dominant matrix for $\lambda > 0$. Hence

$$G_x(\lambda) := \frac{(M\mathbf{1})_x - 1}{\lambda} = \frac{1}{r} + \frac{a/r}{\lambda + h}, \quad G_y(\lambda) = \frac{1}{s} + \frac{c/s}{\lambda + h}. \quad (16)$$

Select a row with positive *source* entrance and write it as $G = u + v/(\lambda + h)$, $v > 0$. If $G_j = G(\lambda_j)$, then

$$\mathcal{R} = \frac{G_1 - G_2}{G_2 - G_3} \frac{\lambda_3 - \lambda_2}{\lambda_2 - \lambda_1} = \frac{\lambda_3 + h}{\lambda_1 + h} > 1, \quad h = \frac{\lambda_3 - \mathcal{R}\lambda_1}{\mathcal{R} - 1}. \quad (17)$$

For each row set

$$v_i = (G_i(\lambda_1) - G_i(\lambda_2)) \frac{(\lambda_1 + h)(\lambda_2 + h)}{\lambda_2 - \lambda_1}, \quad u_i = G_i(\lambda_1) - \frac{v_i}{\lambda_1 + h}. \quad (18)$$

Then $r = 1/u_x$, $s = 1/u_y$, $a = v_x/u_x$, and $c = v_y/u_y$. If $a_0 > 0$, use

$$d = -rM_{xy}(\lambda_1)(\lambda_1 + h)/a, \quad b = h - d. \quad (19)$$

Otherwise $c_0 > 0$ and use $b = -sM_{yx}(\lambda_1)(\lambda_1 + h)/c$, $d = h - b$. Keep the source-selected branch throughout its neighborhood. No formula divides by an entrance vanishing at a leaf.

All denominators are nonzero at the source and remain so locally. The inverse is therefore locally Lipschitz as a function of the finitely many transform entries. Those entries are bounded continuous tests of row laws and are Lipschitz in TV. Two-state data have both G rows constant and cannot approach the selected nonzero source differences. At a complete source the recovered rates stay bounded away from zero, and the normalized stationary solution and (2) are smooth. This proves the entropy statements as well. $\square$

No trace or rate cap is assumed in this reconstruction. The result does not assert that three arguments are the fewest possible, or that differentiation of noisy waiting-time laws is continuous.

Proposition 3 (A fixed-trace three-state boundary). *For $0 < k < e < 1/2$, let*

$$Q(e, k) = \begin{pmatrix} -1-e & 1 & e \\ 1 & -2+k & 1-k \\ k & 1-e & -1+e-k \end{pmatrix}. \quad (20)$$

These generators have trace -4 , off-diagonal rates at most one, and observed rates $r = s = 1$. Each exact kernel uniquely determines its generator in $\mathcal{C}_{\leq 3}$. With $Z = 3 + e + 2k - e^2 - ek - k^2$,

$$\sigma(Q(e, k)) = \frac{(e-k)(1-e-k)}{Z} \log \frac{e(1-e)}{k(1-k)}. \quad (21)$$

Their row distance from the unit tree $Q(0, 0)$ is below $4e$. Along $k = \exp(-1/e^2)$, as $e \downarrow 0$,

$$Q(e, k) \longrightarrow Q(0, 0), \quad e\sigma(Q(e, k)) \longrightarrow \frac{1}{3}. \quad (22)$$

Every positive observation neighborhood of the tree has unbounded entropy even with the stated trace and rate cap.

Proof. The unnormalized stationary weights are

$$(w_x, w_y, w_h) = (1 - e + 2k - k^2, 1 + k - e^2, 1 - k + 2e - ek). \quad (23)$$

They sum to Z and solve $wQ = 0$. Along the cycle $x \rightarrow h \rightarrow y \rightarrow x$, each edge current is $(e-k)(1-e-k)/Z$, and the cycle affinity is the logarithm in (21). This proves the entropy identity. The source's mean next-mark times from (x, y, h) are $(1, 2, 3)$. Common/excess-clock coupling has separation hazard at most $e+k$, hence row TV is at most $2(e+k) < 4e$. In (21), $Z \rightarrow 3$ and $\log(1/k) = 1/e^2$, giving (22). Alternatively, first fix any sufficiently small $e > 0$, then let $k \downarrow 0$: the logarithmic coefficient tends to $e(1-e)/(3+e-e^2) > 0$. This proves an infinite supremum in a fixed neighborhood. The two limiting statements are distinct. $\square$

More generally, at any noncomplete $Q_0 \in \mathcal{C}_N$, open a missing pair at rates e and $\exp(-1/e^2)$ and subtract their sum from an existing positive unobserved ordered rate. For small e this preserves connected reciprocal support, observed rates, trace, and the original maximum rate. Stationarity varies continuously, so the new edge contributes asymptotically π_i^0/e . Thus generator-level entropy is locally upper bounded exactly at complete sources. A complete generator nevertheless need not give observational upper boundedness when more hidden states are present.

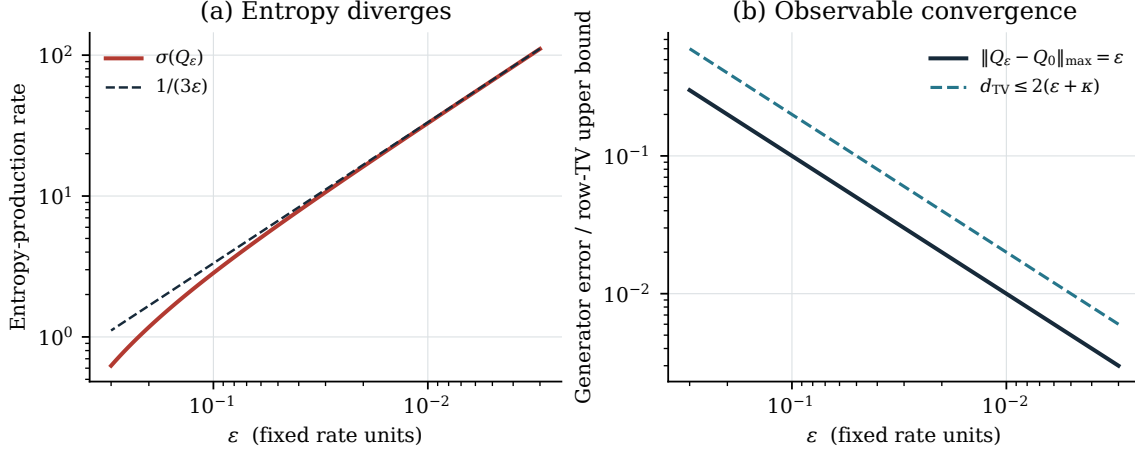


FIG. 2. The deterministic family (20) with $k = \exp(-1/e^2)$; the figure uses $\epsilon = e$ and $\kappa = k$. The rates converge to the unit tree and the analytic row-TV upper bound $2(e + k)$ tends to zero, while entropy diverges with asymptote $1/(3e)$. The plotted TV line is a bound, not a measured distance. All finite-parameter models retain strictly positive reverse rates, observed rates one, trace -4 , and off-diagonal rates at most one. Arbitrary-precision evaluation avoids replacing the small reverse rate by zero.

V. A SHARP CRITERION FOR FINITE LOCAL UPPER CEILINGS

For a source Q_0 , denote by $\mathcal{C}_N(Q_0; r, s, \tau)$ the subclass fixing its two observed rates and $\tau = -\text{tr } Q_0$. No further individual rate restriction is implicit.

Theorem 4 (Fixed-count local criterion). *For $Q_0 \in \mathcal{C}_N$ with one observed reverse pair and unknown competitor topology, a finite local entropy upper ceiling exists in $\mathcal{C}_N$ if and only if*

$$N = 2 \quad \text{or} \quad N = 3 \text{ and } Q_0 \text{ has complete support.} \quad (24)$$

The same criterion holds in $\mathcal{C}_N(Q_0; r, s, \tau)$ and in the product weak topology of joint row laws.

The large- N part follows from a more general constructive statement. For a finite set of resolved marks, extend (6) by $R_I = e_{\text{target}(I)}^T$ and $B_J = q_J e_{\text{source}(J)}$, with $T = Q - \sum_J E_J$. Extend (8) by maximizing over all previous marks and summing over all next marks.

Lemma 5 (Two fully hidden states). *Allow any nonempty reverse-closed set of individually resolved observed edges on a fixed state set. Suppose two distinct states are absent from every observed endpoint. For every source Q_0 , every $\eta > 0$, and every finite L , there is a complete bidirected generator Q' with the same state count, every observed rate, and trace, such that*

$$d(\mathcal{O}(Q'), \mathcal{O}(Q_0)) < \eta, \quad \sigma(Q') > L. \quad (25)$$

Every off-diagonal rate of the constructed models is at most $\tau = -\text{tr } Q_0$. No minimality assumption is needed.

Proof. Let h, k be fully hidden and $\mathcal{U}$ the unobserved ordered pairs. Their source rate mass $u = \sum_{(i,j) \in \mathcal{U}} q_{ij}^0$ is positive. Give each such pair weight one except (h, x) , for one observed endpoint x , which receives weight two. Define G by retaining the observed rates, assigning unobserved rate $uw_{ij}/\sum_{\mathcal{U}} w_{ij}$, and normalizing rows. Then G is complete and has the same trace as Q_0 . Put $P_\epsilon = (1 - \epsilon)Q_0 + \epsilon G$. For all sufficiently small $\epsilon > 0$ it is complete and the full external exit rows of h, k differ. Indeed their difference is $(1 - \epsilon)v + \epsilon w$ with $w \neq 0$, so it is nonzero near zero whether v vanishes or not.

Let $T_0 = Q_0 - E$, where E is the sum of marked matrices. All marked matrices, R , and B stay fixed. Positivity of the killed semigroups, Duhamel's identity, and $\int_0^\infty e^{(P_\epsilon - E)t} B \mathbf{1} dt = \mathbf{1}$ give

$$d(\mathcal{O}(P_\epsilon), \mathcal{O}(Q_0)) \leq \frac{\epsilon}{2} \max_I [R(-T_0)^{-1} |G - Q_0| \mathbf{1}]_I. \quad (26)$$

The constant is finite. Appendix D gives an exact hidden-pair similarity orbit of each fixed P_ϵ that preserves its complete marked kernel and approaches a one-way limiting flux with positive opposing flux. Its entropy diverges. First choose ϵ to make (26) smaller than η , then choose a member of this fixed orbit with entropy above L . Similarity retains trace, and nonnegative rates sum to τ , giving the claimed common upper cap. The order of choices does not require any uniform lower bound on the divergence coefficient as $\epsilon \rightarrow 0$. $\square$

Proof of Theorem 4. Two-state chains have zero stationary current and entropy. At a complete three-state source, Theorem 2 gives local entropy regularity, hence a finite ceiling, in either data topology; restricting the class preserves it. Every three-state tree admits the missing-pair construction after Proposition 3. There is a positive unobserved edge to compensate, so observed rates and trace remain fixed. The same coupling used there gives row-TV convergence and divergent entropy. For $N \geq 4$, one observed pair leaves at least two fully hidden states, and Lemma 5 applies. TV convergence implies weak convergence, so these negative cases also exclude a weak neighborhood ceiling. $\square$

The cap supplied by the construction is τ , not every upper cap containing the source. For example, at $N = 4$, trace -12 and $q_{ij} \leq 1$ force all twelve rates to equal one. That restricted class is a singleton despite two fully hidden states. Known sparse topology and positive reverse-rate information can likewise change the conclusion.

TABLE I. Local upper boundedness under unknown bidirected support and fixed state count. The observation column uses joint waiting laws for one resolved pair.

Source	Entrywise-generator neighborhood	Observation neighborhood
$N = 2$	Yes	Yes
$N = 3$, complete	Yes	Yes
$N = 3$, tree	No	No
$N \geq 4$, complete	Yes	No
$N \geq 4$, noncomplete	No	No

VI. FINITE RECORDS AND ADDITIONAL RATE INFORMATION

Let $\mathbb{P}_Q^H$ denote the stationary observed-record law on $[0, H]$, including empty records and both censored ends. The divergent constructions above also give, at their negative-case sources,

$$\sigma(Q_n) \rightarrow \infty, \quad \text{TV}(\mathbb{P}_{Q_n}^H, \mathbb{P}_{Q_0}^H) \rightarrow 0 \quad \text{for each fixed } H > 0. \quad (27)$$

For the sparse sequence this follows from stationary-distribution continuity and common/excess-clock coupling. For the hidden-pair sequence, the completion $P_\epsilon \rightarrow Q_0$ has the same property, and each subsequent similarity preserves the stationary marked-record law exactly. To see the latter, use $\pi' = \pi S$, $S\mathbf{1} = \mathbf{1}$, and $SE_J = E_J S$ in each record density

$$\pi e^{Tt_1} E_{J_1} e^{T(t_2-t_1)} \dots E_{J_n} e^{T(H-t_n)} \mathbf{1}. \quad (28)$$

All similarities cancel, also for $n = 0$. For (20) a quantitative bound is

$$\text{TV}(\mathbb{P}_{Q(e,k)}^H, \mathbb{P}_{Q(0,0)}^H) \leq \frac{(e-k)(5+e-k)}{3Z} + (e+k)H \leq \frac{11e}{18} + 2eH. \quad (29)$$

These statements establish convergence in a distinct experiment, not by an assumed implication from row TV alone.

Corollary 6 (Uniform upper-confidence obstruction). *Suppose a model class contains a sequence (27). Let U be a measurable extended-real function of the stationary record on a fixed $H > 0$, possibly with independent randomization. If, for $0 < \alpha < 1$,*

$$\mathbb{P}_Q^H\{U \geq \sigma(Q)\} \geq 1 - \alpha \quad \text{for every model in the class,} \quad (30)$$

then $\mathbb{P}_{Q_0}^H\{U = +\infty\} \geq 1 - \alpha$.

Proof. For every finite L , eventually $\sigma(Q_n) > L$. Transfer the coverage inequality through TV convergence to obtain $\mathbb{P}_{Q_0}^H\{U > L\} \geq 1 - \alpha$. Let $L \uparrow \infty$ and use continuity from above. Adjoining an independent random seed leaves TV convergence unchanged. $\square$

This is an application of the established confidence-set impossibility mechanism [15], with the required physical sequences supplied here. It concerns uniform finite-sample coverage over the declared class. It does not settle pointwise asymptotics, Bayesian prior restrictions, or a horizon growing jointly with the sequence. We also do not infer a cap-free positive continuity theorem from one stationary finite window; Theorem 2 concerns complete joint waiting laws.

Additional microscopic information can restore a finite upper bound. If each supported rate lies in $[m, M]$ with $m > 0$, then (3) gives

$$\sigma \leq A \log(M/m), \quad A = \sum_i \pi_i \sum_{j \neq i} q_{ij} \leq (N-1)M, \quad (31)$$

and $A \leq \tau$ when trace is fixed. More generally a reciprocal rate-ratio bound $q_{ij}/q_{ji} \leq R_0$ and activity bound $A \leq A_0$ imply $\sigma \leq A_0 \log R_0$. Such microscopic restrictions are consistent with the hypotheses of stronger rate-ratio upper bounds [14]. A ratio bound alone does not suffice: a three-cycle with clockwise rates $2c$ and reverse rates c has $\sigma = c \log 2$ and $A = 3c$. Nor is a compact set of irreducible limiting generators sufficient without controlling newly opening support: Proposition 3 converges to a single irreducible tree and still diverges.

VII. DISCUSSION

The exact fiber and the local upper envelope answer different questions. Theorem 1 provides a fully resolved exceptional fiber, with precisely two microscopic entropy values. Theorem 4 shows that even this finite exact answer, and the unique generators immediately above its threshold, do not give a finite ceiling at positive observational tolerance in the same state-count class. The obstruction persists under an upper rate bound furnished by fixed trace.

The source of this behavior depends on dimension. With one hidden state, bounded Laplace tests recover the physical generator rationally and stably, including at leaves. Entropy becomes singular because an added forward flux and its reverse vanish on very different scales. With two fully hidden states, an exact similarity can hide a one-way limiting flux even when the source has complete support. Table I separates this inverse ambiguity from the generator-level singularity. The construction does not imply that its high-entropy generators converge entrywise to a complete source.

The closest realization results describe positive equivalence classes, invariant polytopes, or finite factorization ambiguity [7–10]. Those frameworks are inputs to our analysis. The specific addition is the exhaustion over reciprocal physical supports with fixed rank-one event maps, followed by an exact calculation of the full entropy image. Likewise, waiting-time inference results and hidden-dissipation bounds [1–3, 5] motivate the observation but do not substitute for this whole-fiber count or the local necessary-and-sufficient criterion. These are precise distinctions from the cited results; they do not constitute a claim that every related realization theorem has been excluded.

Several restrictions are consequential. Unknown support allows new reciprocal edges; a known tree instead has zero stationary entropy. A state cap excludes nonminimal splitting, and the number of states cannot silently change in the local criterion. The observed marks specify microscopic incidence and retain their absolute amplitudes. Aggregated event labels, uncertain endpoints, multiple channels, odd variables, or time-dependent generators are outside the model. Even here, the full two-parameter theta family and arbitrary five- or six-state entropy fibers remain unclassified. The balanced construction is an existence and exhaustion result for a particular informative family, not an exhaustive small-network census.

VIII. COMPUTATIONAL DETAILS AND REPRODUCIBILITY

The universal claims are established by the proofs in this article. Computation verifies exact identities, admissibility, and the rational entropy enclosures; numerical plots illustrate proved limits. There is no external dataset and no numerical optimization or graph enumeration used as a nonexistence proof.

OpenAI Codex, a GPT-6-based research and coding assistant, was used for literature retrieval, derivation and code preparation, manuscript drafting, and separate mathematical, citation, and reproducibility reviews. The checks below identify the executed validation; agreement between assistant reviews is not treated as independent scientific evidence. Expert human review remains pending.

The supplied package contains the source code and accepted outputs required for these checks. Exact arithmetic uses rational numbers and the real field $\mathbb{Q}(z_*)$, with $\mathbb{Q}(\sqrt{6})$ for the cone. Signs are evaluated in a specified real embedding through rational isolating intervals, not from leading polynomial coefficients. The endpoint checker validates all denominator exclusions, support zeros, positive rates, normalization, stationarity, and ten exact Markov-parameter matrices. Its independent cone and physical-path constructions agree coefficientwise. Appendix C specifies the

logarithm remainder bounds.

The deterministic reproduction driver replays the principal symbolic certificates with output writes redirected, preserving historical artifacts. It independently checks the three-state stationary weights, cycle current, trace, inverse identities, and deterministic high-precision values. The figures use fixed parameter grids; no random seed, optimizer tolerance, or graph-sampling distribution enters the publication results. The accompanying manifest records commands, dependency versions, source hashes, and exact checks versus numerical diagnostics. Discovery searches preserved in the research repository are historical leads and are unnecessary for the certificates. No result is claimed formally verified; the repository’s Lean scaffold is not a verification of these theorems.

IX. CONCLUSION

Complete resolved-transition waiting laws can have an exactly two-point microscopic entropy fiber, but finite local entropy upper bounds require stronger conditions. With unknown bidirected support and one observed reverse pair, the fixed-count criterion is sharp: two states or a complete three-state source. Exact inverse reconstruction, state-count information, and a finite upper rate scale therefore do not generally control hidden dissipation when arbitrarily small reverse rates remain admissible.

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Data availability. All inputs are specified analytic generators. The accompanying reproducibility package supplies the LaTeX sources, figure scripts, exact-check scripts, and machine-readable outputs. The authors should select the final public archive or other journal-compliant access arrangement before submission; no public deposit is asserted in this draft.

Appendix A: Observation equivalence and minimum order

A realization (R, T, B) is minimal when the columns of $T^j B$, $j \geq 0$, span its state space and $RT^j v = 0$ for every $j \geq 0$ implies $v = 0$. A sufficiently large block Hankel matrix $(RT^{i+j} B)_{i,j}$ then has rank equal to the state dimension. Any competing realization factors this same matrix through its own state space, proving the minimum-order assertion without a pole-count shortcut.

Lemma 7 (Finite certificate and pinned similarity). *Two realizations of dimensions n, n' have the same all-time kernel if and only if their Markov parameters $RT^j B$ agree for $0 \leq j < n + n'$. Equivalent minimal realizations have the same dimension and a unique invertible similarity. For the resolved pair of Sec. II, it has the form*

$$S = \text{diag}(I_2, U), \quad U\mathbf{1} = \mathbf{1}, \quad Q' = S^{-1}QS. \quad (\text{A1})$$

Conversely, every such similarity yielding an admissible generator preserves the complete kernel. Entries of U need not be nonnegative.

Proof. Apply Cayley–Hamilton to $\text{diag}(T, T')$: zero differences of the first $n + n'$ parameters force all subsequent differences to vanish. The convergent exponential series then gives all-time equality; the converse follows by differentiation. For minimal realizations define, on reachable vectors,

$$S \left(\sum_j T'^j B' v_j \right) = \sum_j T^j B v_j. \quad (\text{A2})$$

Equality of parameters and observability make this well defined; reversing the argument gives its inverse. Controllability gives $SB' = B$, $ST' = TS$, and $RS = R'$, and uniqueness. The zero-time data give $q_{xy} = \psi_{-,+}(0)$ and $q_{yx} = \psi_{+,-}(0)$. The common R, B therefore pin every visible row and column of S , giving its block form. Since $T\mathbf{1} = -B\mathbf{1}$ and T is invertible, $TS\mathbf{1} = ST'\mathbf{1} = -SB\mathbf{1} = T\mathbf{1}$ implies $S\mathbf{1} = \mathbf{1}$. The observed matrices act only on the fixed visible coordinates, hence $Q' = S^{-1}QS$. Direct substitution proves the converse. $\square$

This is the realization-basis argument specialized to the event maps; its positive-model antecedents are discussed in Sec. I [7, 8].

For the balanced source, the hidden blocks are

$$X = \begin{pmatrix} 2 & 0 & 8 \\ 0 & 7 & 11 \end{pmatrix}, \quad Y = \begin{pmatrix} 3 & 0 \\ 0 & 6 \\ z & z \end{pmatrix}, \quad H = \begin{pmatrix} -7 & 4 & 0 \\ 5 & -11 & 0 \\ 0 & 0 & -2z \end{pmatrix}. \quad (\text{A3})$$

Writing Y_j for its columns, direct determinants give

$$\det(Y_1, Y_2, HY_1) = -9z(4z - 9), \quad (\text{A4})$$

$$\det(Y_1, Y_2, HY_2) = -18z(2z - 3), \quad (\text{A5})$$

$$\det \begin{pmatrix} X_{1,:} \\ X_{2,:} \\ (XH)_{2,:} \end{pmatrix} = -14(22z + 19). \quad (\text{A6})$$

For $z > 0$ the first two cannot both vanish and the last never vanishes. Thus the hidden transfer has reachable and observable order three. The visible plane is directly selected by R and B . Subtracting its contributions in successive powers of T leaves the hidden reachability and observability spaces just computed, so the full order is five. This includes both hidden-pole coincidences and $z = 2$. The cap forces every competitor to have exactly five states and be minimal.

The observations also determine the visible block A : their zero-time values give the observed rates and first derivatives give the killed visible diagonals. Write $A_{\text{kill}} = A - (E_+ + E_-)_{VV}$. After row permutation and division by observed rates they determine the visible resolvent $K(\lambda)$. Its Schur complement determines

$$X(\lambda I - H)^{-1}Y = \lambda I - A_{\text{kill}} - K(\lambda)^{-1}. \quad (\text{A7})$$

This is a matrix-valued constraint shared by all competitors. Treating its entries as unrelated scalar phase-type fits would lose it.

Appendix B: Global support exhaustion and the invariant triangle

1. Modal data and disconnected hidden supports

Let $\rho = \sqrt{6}$, $\alpha = 9 - 2\rho$, $\beta = 9 + 2\rho$, $r_y = 4(\rho - 1)/5$, and set

$$D = \text{diag}(-\alpha, -\beta, -2z), \quad X_m = (p, q, v), \quad Y_m = \begin{pmatrix} r \\ f \\ w \end{pmatrix}, \quad (\text{B1})$$

$$p = \begin{pmatrix} 3 + \rho/2 \\ 35\rho/8 \end{pmatrix}, \quad q = \begin{pmatrix} 3 - \rho/2 \\ -35\rho/8 \end{pmatrix}, \quad v = \begin{pmatrix} 8 \\ 11 \end{pmatrix}, \quad (\text{B2})$$

$$r = (1, r_y), \quad f = (1, -4(\rho + 1)/5), \quad w = (z, z). \quad (\text{B3})$$

Direct multiplication yields

$$X(\lambda I - H)^{-1}Y = \frac{pr}{\lambda + \alpha} + \frac{qf}{\lambda + \beta} + \frac{vw}{\lambda + 2z}. \quad (\text{B4})$$

The residues pr, vw are strictly positive and rank one; qf has negative off-diagonal entries. At $z \geq z_*$ the three poles are distinct and $2z > \beta$.

A disconnected bidirected hidden support on three states consists of three singletons or a pair and a singleton. Singletons have nonnegative outer-product residues, excluding three singletons and a singleton at β . A singleton at α leaves a pair with dominant eigenvalue $-\beta$; its Perron transfer residue is nonnegative, again impossible. The singleton must carry $-2z$.

The remaining pair has zeroth response $pr + qf = \text{diag}(6, 42)$. Nonnegativity and reciprocal visible incidence force its two states to attach separately to x and y : a state attached to both would give positive off-diagonal response. Write entrances x_1, x_2 , exits y_1, y_2 , and internal rates h_0, k_0 . Zeroth and first moments, with row sums, give

$$x_1 y_1 = 6, \quad x_2 y_2 = 42, \quad y_1 + h_0 = 7, \quad y_2 + k_0 = 11, \quad (\text{B5})$$

$$x_1 h_0 y_2 = 48, \quad x_2 k_0 y_1 = 105. \quad (\text{B6})$$

Hence $h_0 k_0 = 20$ and $19h_0 - h_0 k_0 = 56$, giving $h_0 = 4, k_0 = 5, y_1 = 3, y_2 = 6, x_1 = 2, x_2 = 7$. The singleton's residue and exit sum give entrances $(8, 11)^T$ and exits (z, z) . Thus this branch contains only the source.

2. All connected hidden supports give finite triangles

Throughout the connected-case argument take $z \geq 21/2$, so $2z > \beta$ and the modal spectrum is distinct. For a connected hidden support, minimality gives an invertible real V with $H' = V^{-1}DV$ Metzler irreducible, $X' = X_m V \geq 0$, and $Y' = V^{-1}Y_m \geq 0$. The right Perron eigenvector $V^{-1}e_1$ at $-\alpha$ is positive: its sign is fixed by $X'V^{-1}e_1 = p > 0$. The left eigenvector $e_1^T V$ is then positive as well, since its product with that right eigenvector is one. Every column of V has positive first coordinate. Sectioning $V\mathbb{R}_+^3$ at that coordinate equal to one gives a finite nondegenerate triangle with the origin strictly inside. Positively rescale its columns to $(1, u_i, v_i)^T$.

After rescaling time by $\beta - \alpha$, the section is invariant under

$$(u, v) \mapsto (e^{-t}u, e^{-kt}v), \quad k = \frac{2z - 9 + 2\rho}{4\rho}. \quad (\text{B7})$$

It contains inputs $(1, A)$ and $(-m, B_0)$ and lies in an output wedge, where

$$A = z, \quad B_0 = z/r_y, \quad m = (7 + 2\rho)/5, \quad (\text{B8})$$

$$L = (3 - \rho/2)/8, \quad R_w = 35\rho/88, \quad v \geq L(-m - u), \quad v \geq R_w(u - 1). \quad (\text{B9})$$

These are direct consequences of $Y' \geq 0$, $X' \geq 0$, and semigroup positivity. No sign constraint on V itself was imposed. A ray with zero first coordinate is impossible in this irreducible branch; the disconnected branch was already treated separately.

Any wedge point of nonpositive height has abscissa between $-m$ and 1. A triangle with only one positive-height vertex could not contain both positive-height inputs: the same vertex would have to extend beyond both ends of that interval. The interior origin forces a negative-height vertex. Thus the vertices can be labeled

$$(-\ell, P), \quad (c, Q_v), \quad (d, -W), \quad \ell \geq m, \quad c \geq 1, \quad -m < d < 1, \quad P, Q_v, W > 0. \quad (\text{B10})$$

Put $a = k - 1$. The output, lower-face invariance, upper containment, and lower containment conditions include

$$P \geq L(\ell - m), \quad W \leq \min\{L(m + d), R_w(1 - d)\} < 1, \quad (\text{B11})$$

$$(a\ell + kd)P \leq \ell W, \quad (ac - kd)Q_v \leq cW, \quad (\text{B12})$$

$$B_0 \leq H_c := \frac{(c + m)P + (\ell - m)Q_v}{\ell + c}, \quad (\text{B13})$$

$$A \leq \frac{(c - 1)P + (\ell + 1)Q_v}{\ell + c}, \quad (\text{B14})$$

$$(Q_v + W)(1 - d) \leq (A + W)(c - d), \quad (\text{B15})$$

$$(P + W)(d + m) \leq (B_0 + W)(\ell + d). \quad (\text{B16})$$

For example, differentiating the lower-left supporting line along (B7) at $(-\ell, P)$ gives the first inequality (B12). All displayed geometric denominators are positive.

We now restrict to $z \geq 21/2$, sufficient for the threshold and above. If $d < 0$, the reflected use of (B11), (B12), (B14), and (B16) gives

$$A < \frac{L}{a} \left[m + \frac{B_0 + 1}{R_w} \right] < \frac{25/8 + (25/18)(z + 1)}{z - 7}. \quad (\text{B17})$$

The latter is smaller than z for $z \geq 10$, because its reversed difference has numerator $72(z - 10)^2 + 836(z - 10) + 835 > 0$. Here we used $m < 5/2$, $L < 1/4$, $R_w > 9/10$, $B_0 < z$, and $a > (z - 7)/5$. Thus $d \geq 0$. Now $a > 7/10$ and (B12) give $P < 10/7 < 2 < B_0 < A$. Equations (B13)–(B14) imply $Q_v \geq A$ and $\ell > m$.

Write $\epsilon = 1 - d$. Since $aQ_v - W > 0$, the second inequality (B12) and $c \geq 1$ imply

$$\epsilon \leq \frac{A + W}{kA} \leq \frac{A + R_w\epsilon}{kA}, \quad \epsilon \leq \frac{1}{k - R_w/A}. \quad (\text{B18})$$

The wedge apex has

$$d_0 = \frac{R_w - Lm}{L + R_w} > 0, \quad 1 - d_0 = \frac{L(m+1)}{L + R_w}. \quad (\text{B19})$$

At $z = 21/2$, the last fraction minus the bound in (B18) exceeds 0.014259 by rational evaluation with $\rho^2 = 6$ and $\rho > 0$. Since $k - R_w/z$ increases, $d > d_0$ throughout the range. Therefore the active bottom constraint is $W \leq R_w \epsilon$.

3. Global envelope and its unique equality case

At fixed d, W , the left inequalities imply

$$L(\ell - m) \leq g(\ell) := \frac{\ell W}{a\ell + kd}. \quad (\text{B20})$$

Since $d > 0$, g increases strictly and is concave. There is one intersection $\ell_* = m + t$, $t > 0$, with

$$P_* = Lt = g(m + t), \quad Lt[a(m + t) + kd] = (m + t)W. \quad (\text{B21})$$

The corresponding quadratic in t has positive leading coefficient and negative constant. All admissible ℓ, P satisfy $\ell \leq \ell_*$, $P \leq P_*$. In (B13), both partial derivatives in P and ℓ are strictly positive when $Q_v > P$; explicitly they are $(c + m)/(\ell + c)$ and $(c + m)(Q_v - P)/(\ell + c)^2$. Replacing the left vertex by its maximum therefore bounds every feasible triangle, even if a relaxed triangle violates an unused constraint.

For the right vertex, (B15) bounds Q_v by an increasing line, $Q_v \leq (A + W)(c - d)/\epsilon - W$. On $ac - kd > 0$, (B12) gives the decreasing hyperbola $Q_v \leq cW/(ac - kd)$. The relevant intersection is

$$c_* = \frac{k}{a} \left[d + \frac{W\epsilon}{A + W} \right], \quad Q_*^v = \frac{Ad + W}{a\epsilon}. \quad (\text{B22})$$

It lies strictly beyond the pole kd/a . The other algebraic intersection is $c = d < 1$ and is outside the domain. If $c_* < 1$, the hyperbola at $c = 1$ is below A , contradicting $Q_v \geq A$. Along the line, H_c increases in c , with derivative numerator $(\ell_* - m)[(A + W)(\ell_* + d)/\epsilon + W + P_*] > 0$. Along the hyperbola it decreases whenever $Q_v > P_*$; a point with $Q_v \leq P_*$ cannot attain $B_0 > 2$. Thus the unique maximal right choice is (B22).

Set $r_c = c_*/Q_*^v = k\epsilon/(A + W)$. The resulting envelope is

$$H_* = \frac{t[Lm + (1 + Lr_c)Q_*^v]}{m + t + r_c Q_*^v}. \quad (\text{B23})$$

As W increases, t and Q_*^v increase strictly and r_c decreases strictly. The partial derivatives of (B23) with respect to t, Q_*^v are positive; its derivative in r_c has sign $Lt - Q_*^v < 0$. These signs persist up to $W = R_w \epsilon$, since $Lt < 2 < A \leq Q_*^v$. Hence the final maximum uses $W = R_w \epsilon$.

Let $J = am + k$. Equations (B21) and $W = R_w\epsilon$ give

$$\epsilon = \frac{Lt(J + at)}{R_w m + (kL + R_w)t}, \quad H_* - B_0 = \frac{f_0 + f_1 t + f_2 t^2}{(J + at)^2}, \quad (\text{B24})$$

where

$$f_0 = AR_w m / L - B_0 J^2, \quad (\text{B25})$$

$$f_1 = A(R_w / L + k - am) + LJ^2 + R_w J - 2B_0 J a, \quad (\text{B26})$$

$$f_2 = a[LJ + R_w - A(1 + Lk/R_w) - B_0 a]. \quad (\text{B27})$$

These identities follow by direct rational substitution, with no optimization assumptions. The coefficients obey

$$f_0 = -\frac{19 + 9\rho}{100} z \left(z^2 - 11z + \frac{211}{44} \right), \quad (\text{B28})$$

$$f_1^2 - 4f_0 f_2 = -\left(\frac{11}{1200} + \frac{11\rho}{4200} \right) \left(z + \frac{19}{22} \right)^2 \frac{\mathcal{P}(z)}{352}. \quad (\text{B29})$$

Descartes' rule, $\mathcal{P}(0) < 0$, and $\mathcal{P}(z) \rightarrow +\infty$ as $z \rightarrow +\infty$ give a unique positive root. Rational evaluation isolates it in $(10.55714966866, 10.55714966867)$. The larger root of the quadratic in (B28) is below this interval. Thus for $z > z_*$, $f_0 < 0$ and the discriminant is negative; the quadratic in (B24) is strictly negative everywhere. This excludes every connected support.

At $z = z_*$, exact root-interval evaluation gives

$$-0.520929 < f_0 < -0.520928, \quad 5.496508 < f_1 < 5.496509, \quad -14.498918 < f_2 < -14.498917. \quad (\text{B30})$$

The quadratic is nonpositive and has its sole zero at $t_* = -f_1/(2f_2) > 0$. Feasibility requires equality throughout the chain $B_0 \leq H_c \leq H_* \leq B_0$. Strictness of every maximization fixes the unique vertices through (B21), (B22), and (B24). The left and bottom vertices lie on the left and right output faces, respectively. Input $(1, A)$ lies on the lower-right edge, input $(-m, B_0)$ on the upper edge, and both upper-vertex lower-face derivatives vanish. These paired input/output zeros eliminate one visible pair at each of two vertices; the two tangencies eliminate both directions between the upper vertices. The resulting support is precisely the hidden path specified below.

The unique triangle gives at most one normalized generator, not merely one cone. For its vertex matrix V , put $\bar{H} = V^{-1}DV$, $\bar{Y} = V^{-1}Y_m$, $\bar{X} = X_m V$. Column rescaling is fixed uniquely by

$$w = -\bar{H}^{-1}\bar{Y}\mathbf{1}, \quad W_d = \text{diag}(w), \quad H' = W_d^{-1}\bar{H}W_d, \quad X' = \bar{X}W_d, \quad Y' = W_d^{-1}\bar{Y}. \quad (\text{B31})$$

All hidden eigenvalues are nonzero. Existence, positive scaling, and entropy separation are supplied independently in Appendix C; they do not follow merely from equality in a necessary bound.

Appendix C: Exact endpoint construction and entropy certificate

An explicit rational definition avoids using rounded rates. In the following equations evaluate all expressions at $z = z_*$. Define

$$C = \begin{pmatrix} 3 & 0 & 1 \\ 0 & 6 & 1 \\ z & z & 1 \end{pmatrix}, \quad D_c = C^{-1}HC, \quad (\text{C1})$$

where H is (A3). Use one-based indices and set

$$(a_1, a_2, b_1, b_2, c_1, c_2) = ((D_c)_{22}, (D_c)_{21}, (D_c)_{12}, (D_c)_{11}, (D_c)_{32}, (D_c)_{31}). \quad (\text{C2})$$

Let

$$a_p(z) = 100z^4 - 840z^3 + 3741z^2 - 6867z + 3978, \quad (\text{C3})$$

$$b_p(z) = -408z^4 - 2916z^3 + 8766z^2 + 3105z - 12474, \quad (\text{C4})$$

$$c_p(z) = 2412z^4 - 5724z^3 + 891z^2 + 3402z. \quad (\text{C5})$$

They satisfy the independent physical-path discriminant identity

$$b_p^2 - 4a_p c_p = -567(z-2)^2(2z-3)^2 \mathcal{P}(z). \quad (\text{C6})$$

Set

$$u = -\frac{b_p}{2a_p}, \quad v = -\frac{u(a_2u + c_2)}{u^2 + a_1u + c_1}, \quad (\text{C7})$$

$$p_x = (42 + 11z)/18, \quad p_y = 11z/18, \quad q_x = 8z/10, \quad q_y = (6 + 8z)/10, \quad (\text{C8})$$

$$c'_p = \frac{p_x - u}{p_y}, \quad d'_p = \frac{q_y - v}{q_x}, \quad A_p = \frac{u + c'_p v}{1 - c'_p d'_p}, \quad B_p = \frac{v + d'_p u}{1 - c'_p d'_p}. \quad (\text{C9})$$

The primed c'_p, d'_p are chart coordinates, not the polynomial $c_p(z)$. Finally define

$$C' = \begin{pmatrix} 0 & u & 1 \\ v & 0 & 1 \\ B_p & A_p & 1 \end{pmatrix}, \quad U = C(C')^{-1}, \quad S = \text{diag}(I_2, U), \quad Q_* = S^{-1}Q(z_*)S. \quad (\text{C10})$$

The hidden order is (a, b, c) : a y -only endpoint, an x -only center, and a shared endpoint. The seven reciprocal edges are $xy, xb, xc, ya, yc, ab, bc$. All other off-diagonals are exactly zero. For

orientation,

$$Q_* \simeq \begin{pmatrix} -11 & 1 & 0 & 1.86553087 & 8.13446913 \\ 2 & -20 & 6.92248992 & 0 & 11.07751008 \\ 0 & 6.22824145 & -11.27299011 & 5.04474866 & 0 \\ 2.77737298 & 0 & 3.73491843 & -6.73129745 & 0.21900604 \\ 10.48328058 & 10.38263173 & 0 & 0.24409946 & -21.11001177 \end{pmatrix}. \quad (\text{C11})$$

This rounded matrix is not the equality certificate.

The following is a finite, reproducible algebraic admissibility check. Work in the real field $\mathbb{Q}(z_*)$ with $\mathcal{P}(z_*) = 0$, reducing each element to a polynomial of degree at most three. Isolate the positive root in

$$\left[\frac{6329583361674357233}{599554194108037085}, \frac{5875360061222171594}{556529010729195633} \right]. \quad (\text{C12})$$

The interval has width below 10^{-35} ; opposite polynomial signs and unique positivity select the embedding. Rational interval evaluation verifies that $\det C = 18 - 9z$, a_p , $u^2 + a_1u + c_1$, p_y , q_x , $1 - c'_p d'_p$, and $\det C'$ are all nonzero. Reduce (C10) exactly: $U\mathbf{1} = \mathbf{1}$, row sums vanish, the six absent off-diagonal entries are zero, and the fourteen remaining off-diagonals are positive. Their enclosures lie strictly between $1/5$ and 12 . The stated seven-edge graph is connected. This suffices for admissibility. The package supplies every reduced coefficient, making these finite checks repeatable without solving a new realization problem.

Compute the unique normalized stationary vectors by exact reduced linear solves. Both are positive and satisfy $\pi_* = \pi S$. Directly verify $RT^j B = RT_*^j B$ for $j = 0, \dots, 9$; Lemma 7 gives all-time equality independently of the displayed similarity. As a separate coordinate check, the equality triangle in Appendix B, normalized by (B31), agrees coefficientwise with (C10) after swapping its last two hidden vertices. This computation uses pairs $a + b\sqrt{6}$ over $\mathbb{Q}(z_*)$; every final $\sqrt{6}$ coefficient cancels.

Entropy separation also needs a certificate, since different generators need not have different entropy. Evaluate stationary currents and positive flux ratios with rational intervals from (C12). To bound a logarithm, reduce its positive rational endpoints by powers of two into $[1, 2]$. For $t = (x - 1)/(x + 1) \in [0, 1/3]$ use

$$\log x = 2 \sum_{j=0}^{n-1} \frac{t^{2j+1}}{2j+1} + R_n, \quad 0 \leq R_n \leq \frac{2t^{2n+1}}{(2n+1)(1-t^2)}, \quad n = 24. \quad (\text{C13})$$

Bound $\log 2$ by the same formula. Interval arithmetic accounts for the signs of powers-of-two shifts,

currents, and products, and sums once per unordered present edge. It gives (12)–(13) and

$$0.002633285729 \leq \sigma(Q_*) - \sigma(Q(z_*)) \leq 0.002633285730. \quad (\text{C14})$$

These are rational enclosures, not precision-based equality tests. Together with Appendix B, they finish the endpoint and supercritical parts of Theorem 1.

Appendix D: An exact entropy-divergent hidden-pair orbit

Take fully hidden states h, k with the same external neighbor set J , and different full external exit rows Y_h, Y_k . Let $m_0 = q_{hk}$, $n_0 = q_{kh}$, and let

$$\lambda_h = m_0 + \sum_{i \in J} Y_{hi}, \quad \lambda_k = n_0 + \sum_{i \in J} Y_{ki} \quad (\text{D1})$$

be their *total* escape rates. Assume either $m_0, n_0 > 0$, or $m_0 = n_0 = 0$ with $\lambda_h = \lambda_k$. The external rows include all other hidden states, not only visible endpoints.

Exchange h, k if necessary so that $r_* := \min_{i \in J} Y_{hi}/Y_{ki} \in (0, 1)$. Such an orientation exists because the rows differ and their supported entries are positive. Embed

$$U_e = \begin{pmatrix} 1-e & e \\ 0 & 1 \end{pmatrix}, \quad 0 \leq e < r_*, \quad (\text{D2})$$

as the hidden block of S_e . The transformed outside entrances and exits are

$$q'_{ih} = (1-e)q_{ih}, \quad q'_{ik} = q_{ik} + eq_{ih}, \quad (\text{D3})$$

$$q'_{hi} = (Y_{hi} - eY_{ki})/(1-e), \quad q'_{ki} = Y_{ki}. \quad (\text{D4})$$

They retain their reciprocal support before r_* . For an adjacent pair,

$$q'_{kh} = n_0(1-e), \quad q'_{hk} = \frac{f(e)}{1-e}, \quad f(e) = m_0 + e(\lambda_k - \lambda_h) - n_0e^2. \quad (\text{D5})$$

Let e_* be the first zero of f in $(0, r_*]$, if present, and otherwise r_* . Then $0 < e_* \leq r_* < 1$, and all generators before e_* retain the original support. For the unjoined equal-escape pair the hidden block is scalar and unchanged; take $e_* = r_*$.

Normalized similarity preserves the exact kernel. The stationary law is explicitly

$$\pi'_i = \pi_i \ (i \notin \{h, k\}), \quad \pi'_h = (1-e)\pi_h, \quad \pi'_k = \pi_k + e\pi_h. \quad (\text{D6})$$

Every stationary mass has a positive finite limit and all rates remain bounded because $e_* < 1$. If $f(e_*) = 0$, the $h \rightarrow k$ flux vanishes while its reverse tends to $(\pi_k + e_*\pi_h)n_0(1-e_*) > 0$. If

$e_* = r_*$, an external exit $h \rightarrow i$ vanishes while its reverse tends to $\pi_i(1 - r_*)q_{ih} > 0$. This also covers simultaneous zeros. The corresponding nonnegative entropy term diverges, and other terms cannot cancel it. The endpoint itself need not be admitted or have a unique stationary law; (D6) gives the necessary limits before the boundary.

For a minimal realization, equal full external exit rows are impossible: the codimension-one subspace of vectors with equal h, k entries is invariant under T and contains every column of B . Reachability would fail. Thus every pair in a complete minimal realization meets the unequal-row requirement. This observation is used below, while Lemma 5 creates unequal rows explicitly and does not assume minimality.

Appendix E: Complete subcritical coverage, including repeated poles

We give four covering constructions. Each preserves the hidden transfer exactly and yields a pair to which Appendix D applies.

1. Below the slow pole

Take $0 < z < \alpha/2$. Let $d_s = 2(\rho - 1)$. In hidden order (h, k, l) set

$$K_{hl} = -4, \quad K_{kl} = -d_s, \quad K_{lh} = 5, \quad K_{lk} = d_s, \quad K_{hk} = K_{kh} = 0, \quad (\text{E1})$$

and fix diagonals by $K\mathbf{1} = 0$. Under $U = I + \epsilon K$, the derivative of the transformed generator is the commutator with embedded K . The four missing hidden rates have derivatives $(\alpha - 2z)(4, d_s, 5, d_s) > 0$; the missing visible incidences (xk, yh, hy, kx) have derivatives $(8d_s, 55, 4z, d_s z) > 0$. For sufficiently small $\epsilon > 0$, U is invertible, all missing pairs open, and existing positive rates remain positive. The target is complete and minimal, hence has an entropy-divergent hidden pair. This interval includes the path-chart singularity $z = 2$.

2. Between the hidden poles

Take $\alpha/2 \leq z \leq \beta/2$. Put $s_0 = 2z$, $d_s = 2(\rho - 1)$, and $\Delta = 4\rho$. Factor the two relevant residues as

$$Z_{s_0} = \begin{pmatrix} 8 \\ 11 \end{pmatrix} (z, z), \quad Z_\beta = \begin{pmatrix} 2d_s \\ -35 \end{pmatrix} \frac{(3, -24/d_s)}{\Delta}. \quad (\text{E2})$$

Choose midpoints in the nonempty intervals

$$\frac{24}{d_s \Delta z} < t < \frac{4}{d_s}, \quad \frac{3}{\Delta z} < u < \frac{11}{35}. \quad (\text{E3})$$

They are nonempty because $z > \rho/4$ and $z > 35\rho/88$, both already true at $\alpha/2$. For $P_2 = \begin{pmatrix} 1 & 1 \\ -t & u \end{pmatrix}$, use

$$\bar{X} = \begin{pmatrix} 8 - 2d_s t & 8 + 2d_s u \\ 11 + 35t & 11 - 35u \end{pmatrix} > 0, \quad (\text{E4})$$

$$\bar{Y} = \frac{1}{t+u} \begin{pmatrix} uz - 3/\Delta & uz + 24/(d_s \Delta) \\ tz + 3/\Delta & tz - 24/(d_s \Delta) \end{pmatrix} > 0, \quad (\text{E5})$$

$$\bar{H} = \frac{1}{t+u} \begin{pmatrix} -s_0 u - \beta t & u(\beta - s_0) \\ t(\beta - s_0) & -s_0 t - \beta u \end{pmatrix}. \quad (\text{E6})$$

These are obtained by similarity from modes $-s_0, -\beta$. The stable Metzler $\bar{H}$ has two positive off-diagonals for $s_0 < \beta$, and is $-\beta I$ at equality. Normalize by (B31), giving a pair with strictly positive visible incidences. Its exit matrix has rank two because the modal exit rows are a positive row and a mixed-sign row; their independence is preserved.

For the remaining residue write

$$Z_\alpha = cg, \quad c = \begin{pmatrix} 2(d_s + 4) \\ 35 \end{pmatrix}, \quad g = (3, 24/(d_s + 4))/\Delta. \quad (\text{E7})$$

A singleton with entrance $(g\mathbf{1})c/\alpha$ and exit $\alpha g/(g\mathbf{1})$ realizes it. Adjoin it without hidden links to the pair, retaining the source visible block. Hidden row normalization and equality of the transfer at zero give the source's visible hidden-entry totals $(10, 18)^T$, so visible rows also normalize. The full generator is irreducible through its visible links. The pair has unequal external exits, is adjacent for $s_0 < \beta$, and unjoined with equal escapes at $s_0 = \beta$. Appendix D applies in both cases. All transfer identities remain valid when poles coincide: the residues simply add. Thus both endpoints, including $s_0 = \alpha$, are covered without a distinct-pole assumption.

3. An explicit fast opening

Take $\beta/2 \leq z \leq 21/2$. Put $M_s = 2z - 11 > 0$, $L_s = M_s + 4$, and define

$$v_{max} = 7z/33, \quad q = \frac{1}{2}(M_s/5 + 96v_{max}/(zM_s)), \quad v = \frac{1}{2}(zqM_s/96 + v_{max}), \quad (\text{E8})$$

$$u = \frac{1}{2}(zq/24 + 4v/M_s), \quad p = \frac{1}{2}(-4u - zq/6), \quad r = \frac{1}{2}(11v/7 + z/3). \quad (\text{E9})$$

Use hidden off-diagonals $(K_{hk}, K_{hl}, K_{kh}, K_{kl}, K_{lh}, K_{lk}) = (p, q, r, -1, -v, u)$ and $K\mathbf{1} = 0$. Since $L_s M_s \geq 20$ and $11M_s^2 \leq 1100 < 1120$, the midpoint intervals are strict:

$$q > M_s/5, \quad v < 7z/33, \quad zq/24 < u < 4v/M_s, \quad (\text{E10})$$

$$-4u < p < -zq/6, \quad 11v/7 < r < z/3. \quad (\text{E11})$$

The eight missing-rate derivatives, ordered $(xk, yh, hy, hl, kx, kl, lh, lk)$, are

$$(2p + 8u, 7r - 11v, -6p - zq, L_s q - 4, -3r + z, 5q - M_s, L_s v - 5u, 4v - M_s u), \quad (\text{E12})$$

all positive. Thus $I + \epsilon K$ opens a complete realization for small $\epsilon > 0$, with exact observation and subsequent entropy divergence as above. This argument needs neither an exit chart nor an assumption that failure of a first-order opening implies global rigidity.

4. Strict triangles up to the fold

Take $21/2 \leq z < z_*$. Use the quantities of Appendix B and set $t = -f_1/(2f_2)$. On the entire rational interval

$$I = [10.5, 10.55714966867], \quad (\text{E13})$$

rational interval evaluation gives $f_1 > 5$ and $f_2 < -14$. For $z < z_*$, the positive discriminant in (B29) gives

$$f_0 + f_1 t + f_2 t^2 = -\frac{f_1^2 - 4f_0 f_2}{4f_2} > 0. \quad (\text{E14})$$

Define $\epsilon, d, W, \ell, P, c, Q_v$ by (B24), $d = 1 - \epsilon$, $W = R_w \epsilon$, $\ell = m + t$, $P = Lt$, and (B22). Then $B_0 < H_*$ is strict upper-input slack. The remaining geometric signs hold throughout I ; Table II gives conservative rational lower bounds. To reproduce them, enclose ρ between the exact decimal rationals 2.449489742783178098197284 and 2.449489742783178098197285, whose squares straddle six. Shift polynomial coefficients by $z = 21/2 + \xi$, and apply rational interval operations, rounded outward to 24 decimal places after each operation, with $0 \leq \xi \leq 0.05714966867$. No parameter sampling establishes these bounds.

For completeness, the left and right lower-face tangencies imply $p_\ell \ell = kP$ and $q_r c = kQ_v$. Therefore the lower-face margins of the origin are $W - p_\ell d = aP > 0$ and $W + q_r d = aQ_v > 0$. The upper edge is above the origin, so the triangle is nondegenerate with an interior origin. The inward derivatives at the bottom vertex are $a(W + P) > 0$ and $a(W + Q_v) > 0$. The upper-edge inward

TABLE II. Strict rational lower bounds valid on the whole interval (E13), with $p_\ell = (P + W)/(\ell + d)$, $q_r = (Q_v + W)/(c - d)$. The upper-input- B_0 slack is not listed because it vanishes at the fold.

Quantity	Lower bound	Quantity	Lower bound
t	18/100	ϵ	5/100
$1 - \epsilon$	9/10	$c - 1$	1
$L(m + d) - W$	6/10	$Q_v - R_w(c - 1)$	170
$B_0 + W - p_\ell(d + m)$	8	$[(c - 1)P + (\ell + 1)Q_v]/(\ell + c) - A$	100
$W - p_\ell d$	3/100	$W + q_r d$	120
$kW - p_\ell d$	7/100	$kW + q_r d$	120

derivatives at its endpoints have positive numerators $[a\ell + kc]P + \ell Q_v$ and $[k\ell + ac]Q_v + cP$. The two remaining derivatives vanish by tangency. These vertex-face conditions suffice for invariance under the linear flow.

The initial triangle is nonnegative but can have one-way support. Make three small changes for each fixed $z < z_*$. First decrease W while holding d fixed, recomputing (B21) and (B22); all strict margins persist and the bottom output becomes strict. Next decrease ℓ slightly and take P to be the midpoint of $L(\ell - m)$ and $\ell W/(a\ell + kd)$; both left output and left inward inequalities become strict. Finally keep $c = c_*$ and decrease Q_v slightly; since $ac - kd > 0$, both the right inward inequality and the lower containment of $(1, A)$ become strict. All other strict inequalities persist by continuity. Only finitely many conditions are involved, so successive sufficiently small positive choices exist; no uniform size as $z \uparrow z_*$ is claimed.

The final triangle has strictly positive input barycentric coordinates, output values, and all off-diagonal inward derivatives. At vertex j , the derivative of each initially zero barycentric coordinate $i \neq j$ is $(\bar{H})_{ij}$ up to the positive time-rescaling factor; thus strict inward derivatives give positive off-diagonals. Consequently $\bar{X} > 0$, $\bar{Y} > 0$, and every off-diagonal of the stable $\bar{H}$ is positive. The normalization vector in (B31) is positive because $-\bar{H}^{-1} > 0$. Hidden row sums vanish and transfer equality at zero fixes the visible row sums. This is a complete five-state physical generator with the same all-time observation. It is minimal by Appendix A, so its hidden pair meets Appendix D. The four intervals cover all $0 < z < z_*$, completing Theorem 1.

Appendix F: Why one additional state changes the exact answer

Let h be fully hidden in any irreducible generator. Replace it by h_1, h_2 ; keep rates among outside states, split each incoming rate as $\theta_i q_{ih}$ and $(1 - \theta_i)q_{ih}$ with $0 < \theta_i < 1$, and give both clones the original outgoing rates q_{hj} to every outside j . Set internal rates $q_{h_1 h_2} = m > 0$ and $q_{h_2 h_1} = n > 0$, with n fixed. Normalize all rows. Every positive m gives connected reciprocal support with the same observed edges.

For the aggregation matrix C merging the clones, $\tilde{Q}C = CQ$. Because no mark touches h , also $\tilde{T}C = CT$, $\tilde{B} = CB$, and $\tilde{R}C = R$. Therefore $\tilde{\Psi}(t) = \tilde{R}e^{\tilde{T}t}CB = Re^{Tt}B$ exactly. This is strong lumping applied to the killed generator, not just agreement of stationary probabilities.

Put $\lambda = \sum_{j \neq h} q_{hj}$ and $\alpha_c = \sum_{i \neq h} \pi_i \theta_i q_{ih} / \pi_h \in (0, \lambda)$. Outside stationary probabilities remain π_i , and clone probabilities are

$$\tilde{\pi}_{h_1} = \pi_h \frac{\alpha_c + n}{\lambda + m + n}, \quad \tilde{\pi}_{h_2} = \pi_h \frac{\lambda + m - \alpha_c}{\lambda + m + n}. \quad (\text{F1})$$

Both have positive limits as $m \downarrow 0$. The first internal flux tends to zero and the reverse tends to $\pi_h n(\lambda - \alpha_c)/(\lambda + n) > 0$, so entropy diverges while rates remain uniformly bounded above. Applied to $Q(z_*)$, this proves unboundedness with six states. These enlarged realizations are nonminimal for the common marked kernel. This appendix asserts sufficiency of one extra state, not a general necessary state budget.

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